



Day 5: AWS Foundations & Core Networking

AWS TRAINING

DAY 5 OF 6

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DEVOPS ENGINEERING

Stepping into the Cloud — Today we unpack what "The Cloud" actually is, explore the different service and deployment models, and build our first virtual data center using Amazon VPC.

Today's Agenda

What We'll Cover Today

01

Cloud Computing Fundamentals

NIST definition, the 6 advantages, deployment models, and service models

03

Amazon VPC & Networking

VPC components, subnets, security groups, and Network ACLs

02

AWS Global Infrastructure

Regions, Availability Zones, Edge Locations, and the Shared Responsibility Model

04

EC2 Compute & Storage

Instance types, purchase options, EBS vs. EFS



What is Cloud Computing?

The National Institute of Standards and Technology (NIST) defines cloud computing through five essential characteristics that distinguish it from traditional IT infrastructure.

The NIST Five Characteristics

On-Demand Self-Service

Provision compute, storage, and networking resources instantly — no phone calls, no waiting for human approval from the provider.

Broad Network Access

Available over the internet through standard mechanisms and accessible from desktops, laptops, tablets, and mobile phones.

Resource Pooling

A multi-tenant model where the provider's resources serve many customers simultaneously, dynamically assigned based on demand.

Rapid Elasticity

Scale resources up or down almost instantly to match demand — from one server to thousands and back again.

Measured Service

Pay only for what you use. Resource usage is monitored, controlled, and reported — utility-based pricing like electricity.

Why Move to the Cloud?

The 6 Advantages of Cloud Computing

1 Trade Capital Expense for Variable Expense

Pay only when you consume resources. No upfront investment in servers or data centers.

2 Benefit from Massive Economies of Scale

AWS buys hardware in enormous volumes — those savings are passed on to you as lower per-unit pricing.

3 Stop Guessing Capacity

Eliminate costly over-provisioning or under-provisioning. Scale dynamically to match real demand.

4 Increase Speed and Agility

Deploy new resources in minutes, not weeks. Experiment faster, fail cheaper, iterate sooner.

5 Stop Spending on Data Centers

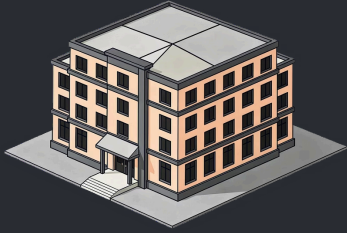
Redirect engineering effort away from "rack and stack" hardware toward innovation that drives business value.

6 Go Global in Minutes

Deploy applications to AWS regions across the globe with just a few clicks — no physical infrastructure required.

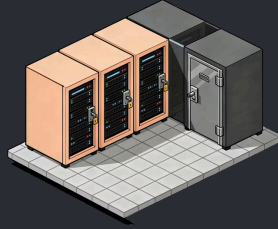
Cloud Deployment Models

Not all cloud deployments look the same. Organizations choose a model based on their security, compliance, and integration requirements.



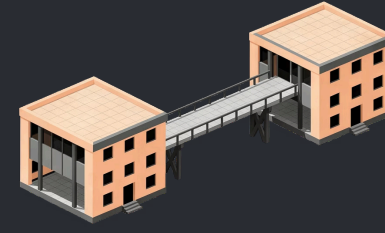
Public Cloud

Resources are owned and operated by a third-party provider like AWS, Azure, or Google Cloud. Open to the public, shared infrastructure, no upfront hardware cost.



Private Cloud (On-Premises)

Resources are used exclusively by a single organization, managed on-site or in a dedicated facility. Highest control and security, but also highest cost.

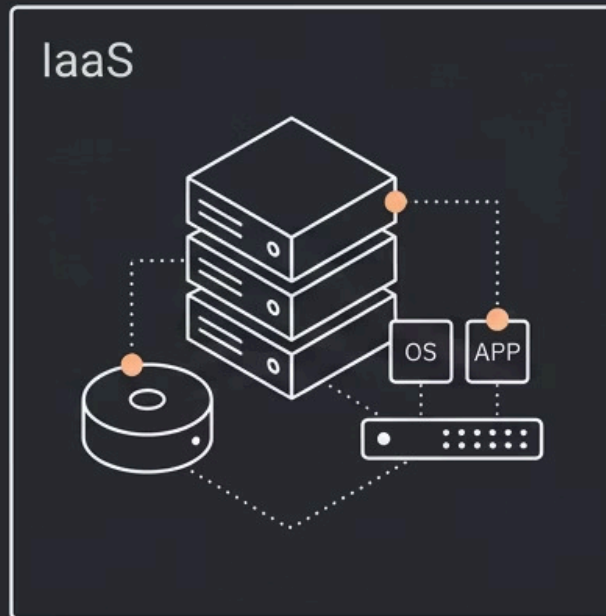


Hybrid Cloud

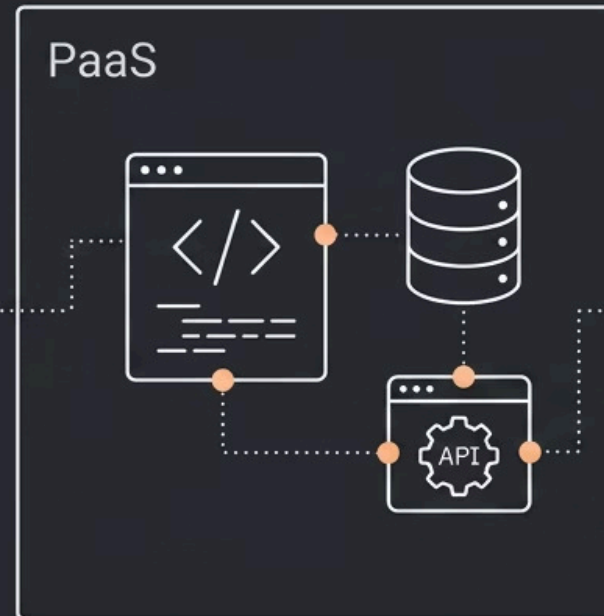
A combination of public and private cloud. Organizations extend on-premises data centers into AWS using **Direct Connect** or a **VPN**, keeping sensitive workloads on-premises.

Service Models: IaaS, PaaS, SaaS

The three cloud service models define **how much you manage** versus how much AWS manages. Think of it as a sliding scale of responsibility.



IaaS (e.g., EC2):
You manage OS,
Apps. Cloud
manages HW,
Virtualization.



**PaaS (e.g.,
Lambda):**
You manage Code.
Cloud manages
OS, Runtime.



**SaaS (e.g., AWS
Console):**
You use it.
Cloud manages
everything.

① Rule of thumb: The higher you go (IaaS → SaaS), the less you configure — but also the less control you have over the underlying infrastructure.



The Shared Responsibility Model

Security is a **joint effort** between AWS and you. Understanding who owns what is fundamental to building secure cloud architectures.

Who Is Responsible for What?

AWS Is Responsible For...

Security OF the Cloud

- Physical data center security
- Underlying hardware and networking
- Global infrastructure (Regions, AZs)
- Hypervisor and managed service security

You Are Responsible For...

Security IN the Cloud

- Your data and encryption
- OS patching on EC2 instances
- Network configuration (VPC, Security Groups)
- IAM users, roles, and policies
- Application-level security

 **Simple Rule:** If you can configure it in the AWS Console, **you are responsible for it.**

AWS Global Infrastructure

AWS runs on a massive global network designed for high availability, fault tolerance, and low-latency content delivery. Understanding this infrastructure is key to architecting resilient applications.



Regions

Geographic locations around the world — **us-east-1**, **eu-central-1**, etc. Each region is fully independent, with its own power, networking, and connectivity. Choose a region based on latency, compliance, and service availability.



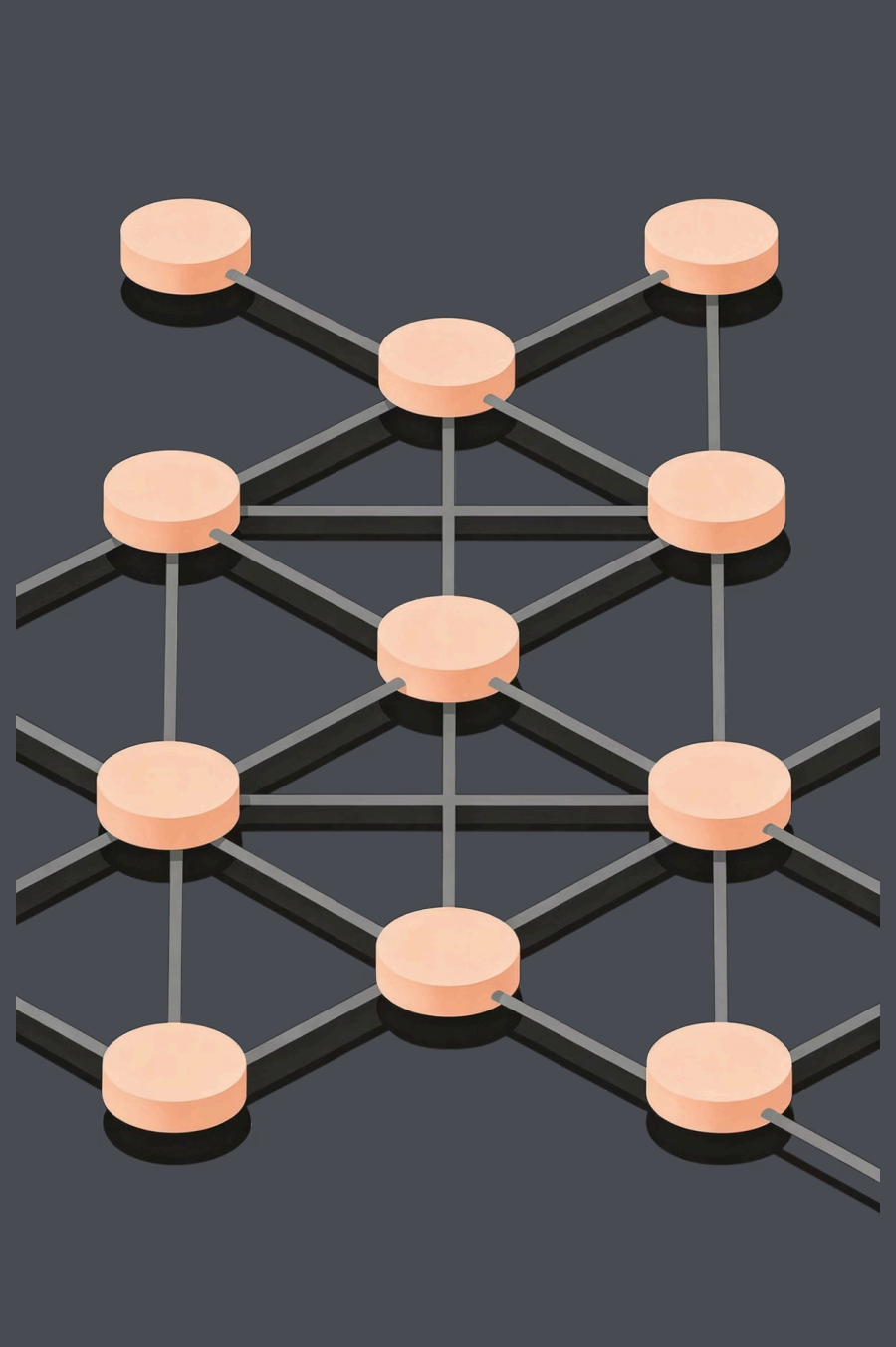
Availability Zones (AZs)

Discrete, physically separate data centers within a Region. Each AZ has redundant power, cooling, and networking. Deploying across multiple AZs protects your application from single-point failures.



Edge Locations

CDN endpoints used by **Amazon CloudFront** to cache and serve content closer to end users. There are hundreds of Edge Locations globally — far more than Regions — minimizing latency for viewers worldwide.



Networking Foundation: Amazon VPC

Your Amazon Virtual Private Cloud (VPC) is your **private, isolated section** of the AWS cloud — your own virtual data center in the sky.

Building Your VPC

Core Concepts

CIDR Block

Defines the IP address range for your entire VPC. Example: 10.0.0.0/16 gives you 65,536 addresses to allocate across subnets.

Public Subnet

A subnet with a route to an **Internet Gateway**. Resources here can be reached from — and can reach — the public internet.

Private Subnet

A subnet with **no direct path** to the internet. Ideal for databases, backend servers, and sensitive resources.

Quick Reference

Component	Scope
VPC	Region-wide
Subnet	Single AZ
Internet Gateway	VPC-wide
Route Table	Subnet-level

 A VPC spans an entire Region. Subnets are scoped to a single Availability Zone.

VPC Core Components



Internet Gateway (IGW)

The front door of your VPC. Attach one to your VPC to enable two-way communication between your public subnets and the internet. Highly available and horizontally scaled by AWS.



Route Tables

A set of rules that determine where network traffic is directed. Every subnet must be associated with a route table. A route of `0.0.0.0/0 → IGW` makes a subnet public.



NAT Gateway

Allows instances in a **private subnet** to initiate outbound connections (e.g., OS updates) without exposing them to inbound internet traffic. Deployed in a public subnet.

💡 Think of it this way: the IGW is the gate for public traffic; the NAT Gateway is a one-way valve for private instances to reach out without being reached.

Network Security: SG vs. NACL

AWS provides two layers of network security within a VPC. Understanding the difference between them is critical for the exam and for real-world architecture.

Security Groups (SG)

- Operates at the **instance (ENI)** level
- **Stateful:** Return traffic is automatically allowed — no need to add an outbound rule for inbound traffic
- Only ALLOW rules — you cannot explicitly deny traffic
- Applied per instance; one instance can have multiple SGs

Network ACLs (NACL)

- Operates at the **subnet** level
- **Stateless:** Inbound and outbound rules must be defined separately and explicitly
- Supports both ALLOW and DENY rules
- Rules are evaluated in numbered order — lower numbers first

⚠ Common trap: If you open port 80 inbound on a NACL, you must also add an outbound rule for the ephemeral port range (1024–65535) or responses will be blocked.

Security Groups



..... **Instance-level (Level)**



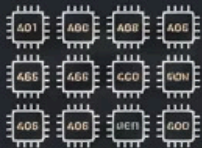
Stateful (State)
Return traffic auto-allowed



..... **Allow rules only (Rules)**

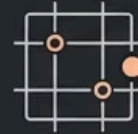


..... **All rules evaluated (Evaluation)**



..... **Fine-grained per-instance control (Use Case)**

Network Access Control Lists (NACLs)



..... **Subnet-level (Level)**



Stateless (State)
Both directions must be defined



..... **Allow & Deny rules (Rules)**



..... **Rules evaluated in order by number (Evaluation)**



..... **Broad subnet-level guardrails (Use Case)**

Compute Power: Amazon EC2

Amazon Elastic Compute Cloud (EC2) is the backbone of AWS compute — virtual servers you can provision in seconds, configure to your exact needs, and pay for only while they're running.



EC2 Key Concepts



AMI

Amazon Machine Image — a pre-configured template containing the OS, application server, and applications. Launch any number of instances from a single AMI.



Instance Types

Each type is optimized for a different workload: **t-series** for burstable general use, **c-series** for compute, **r-series** for memory, **p-series** for GPU tasks.



Free Tier Instances

t2.micro and **t3.micro** are burstable performance instances eligible for the AWS Free Tier — 750 hours/month free for 12 months. Perfect for learning and testing.

EC2 Purchase Options

Choosing the right purchase option can dramatically reduce your compute costs. Match the option to your workload's predictability and flexibility.

On-Demand

Pay by the second. No upfront commitment. Best for **unpredictable or short-term** workloads where you can't interrupt the job.

Reserved Instances

Commit to **1 or 3 years** in exchange for up to **75% discount** over On-Demand pricing. Ideal for steady-state, always-on workloads.

Spot Instances

Bid on spare AWS capacity for up to **90% off** On-Demand pricing. AWS can terminate your instance with a **2-minute notice**. Best for fault-tolerant batch jobs.

Savings Plans

A flexible commitment to a consistent amount of compute usage (\$/hour) over 1–3 years. Applies automatically across instance types, regions, and services like Fargate and Lambda.

 Exam tip: Spot = cheapest but interruptible. Reserved = predictable discount. Savings Plans = flexibility with savings.

EC2 Storage: EBS vs. EFS

EBS — Elastic Block Store

Think: Virtual Hard Drive

- Attached to **one EC2 instance** at a time
- Locked to a **single Availability Zone**
- Persists independently from the instance lifecycle
- Types: gp3 (general purpose), io2 (high IOPS), st1 (throughput)
- Use for: boot volumes, databases, single-server apps

EFS — Elastic File System

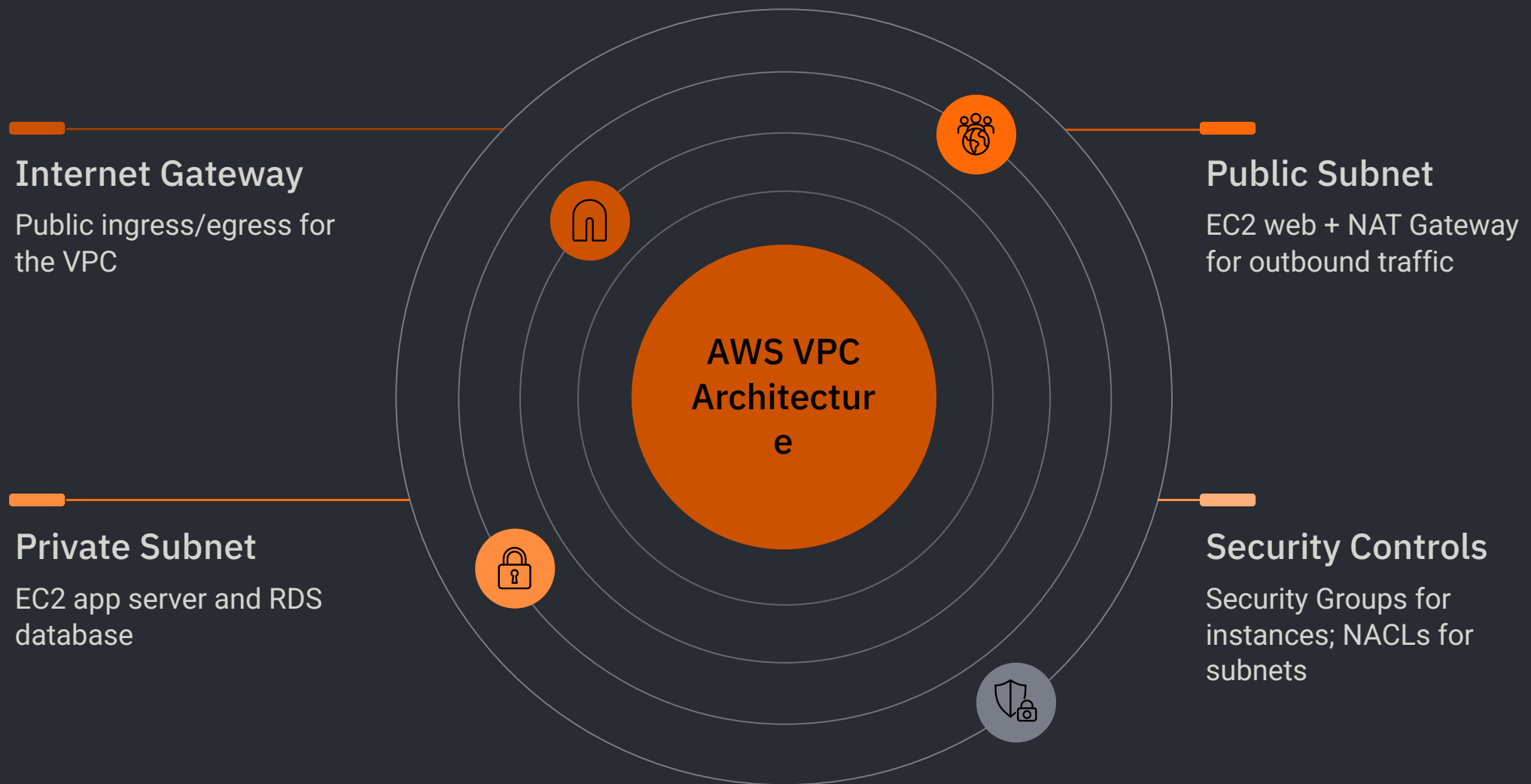
Think: Shared Network Drive

- Mountable by **hundreds of EC2 instances simultaneously**
- Spans **multiple AZs** within a region
- Scales automatically — no capacity planning needed
- Pay for what you store (no pre-provisioning)
- Use for: shared content, CMS, big data workloads

📌 Key distinction: EBS is like a USB drive — one machine at a time, one AZ. EFS is like a network share — many machines, many AZs.

Putting It All Together

Here's how today's concepts connect inside a real AWS architecture: a VPC containing both public and private subnets, secured by Security Groups and NACLs, with EC2 instances accessing the internet through an IGW or NAT Gateway.



Notice how **public-facing resources** sit in the public subnet behind the IGW, while **sensitive backend resources** remain in the private subnet — reachable only from within the VPC.

Day 5: Key Takeaways

→ Cloud = On-Demand + Elastic + Pay-as-You-Go

The NIST five characteristics define cloud computing. The 6 advantages explain *why* organizations are migrating en masse.

→ AWS Infrastructure is Layered

Regions → Availability Zones → Edge Locations. Each layer provides a different type of resilience and performance optimization.

→ VPC is Your Virtual Data Center

CIDR blocks, public/private subnets, IGW, Route Tables, and NAT Gateways are the building blocks of every AWS network architecture.

→ Security is Layered Too

Security Groups (stateful, instance-level) and NACLs (stateless, subnet-level) work together to provide defense-in-depth.

→ EC2 is Flexible and Cost-Optimizable

Choose the right instance type, purchase option, and storage class (EBS vs. EFS) to match your workload and budget.

Quick Knowledge Check

Test your understanding of today's material before we wrap up.

1

Which NIST characteristic enables "pay only for what you use"?

Answer: **Measured Service** — usage is monitored and billed on a utility model.

2

What makes a subnet "public" in a VPC?

Answer: A route in its **Route Table** pointing 0.0.0.0/0 to an **Internet Gateway**.

3

SG vs. NACL — which is stateful?

Answer: **Security Groups** are stateful. NACLs are stateless and require explicit inbound *and* outbound rules.

4

Which EC2 option gives up to 90% savings but can be interrupted?

Answer: **Spot Instances** — AWS can reclaim capacity with a 2-minute warning.



Coming Up Tomorrow — Day 6

We take everything we built today and scale it, store it, accelerate it, and connect it to the internet with managed services.

Day 6 Preview

Auto Scaling & Load Balancing

ASG & ELB — automatically grow and shrink your fleet of EC2 instances and distribute traffic across them for high availability.

Object Storage with S3

Amazon S3 — store, retrieve, and manage virtually unlimited objects in the cloud. Covers storage classes, lifecycle policies, and bucket security.

DNS with Route 53

Amazon Route 53 — connect your custom domain name to your AWS resources using managed DNS, routing policies, and health checks.

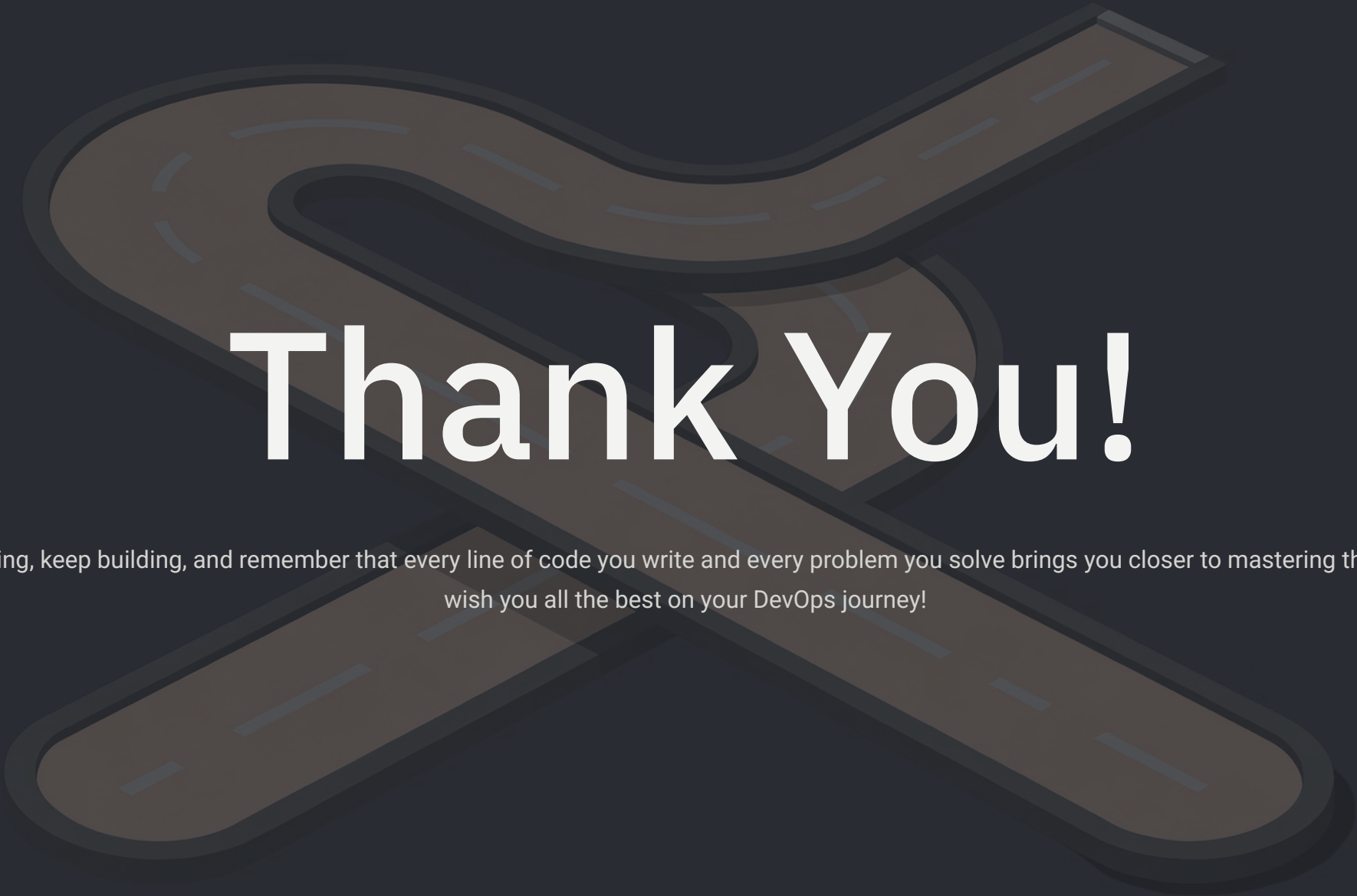
CloudFront CDN

Amazon CloudFront — deliver content from Edge Locations worldwide, dramatically reducing latency for your global users.

Managed Databases with RDS

Amazon RDS — fully managed relational databases (MySQL, PostgreSQL, Aurora, and more) with automated backups, patching, and Multi-AZ failover.

✅ Great work today. You've laid the networking foundation that all future AWS services will build upon. See you tomorrow! 🚀



Thank You!

Keep learning, keep building, and remember that every line of code you write and every problem you solve brings you closer to mastering the cloud. We wish you all the best on your DevOps journey!